

How a gas concentration becomes a receiver decision

Acetonitrile at 90°: one complete numerical calculation

Actual HITRAN parameters, explicit arithmetic and a simulated receiver

What the four numbers in the original table mean

Number	Its role in the calculation
50.5814 $\mu\text{g}/\text{m}^3$	Chosen true enhancement q for this simulated experiment
0.012181 dB	Predicted extra attenuation of the first tone
0.998599	Predicted sample/reference amplitude ratio for that tone
0.1401%	The same amplitude change expressed as a reduction

The last three values follow from q and the molecular absorption model. None is a receiver estimate.

q was chosen from the model's 95% detection limit, before generating the observation. It is not a measured concentration.

q describes an enhancement with a fixed vertical shape

Altitude z	Sample minus reference concentration
0 m	50.5814 $\mu\text{g}/\text{m}^3$
500 m	36.2432 $\mu\text{g}/\text{m}^3$
1500 m	18.6079 $\mu\text{g}/\text{m}^3$
3000 m	6.8454 $\mu\text{g}/\text{m}^3$
5000 m	1.8044 $\mu\text{g}/\text{m}^3$

$$\Delta c(z) = c_{\text{sample}}(z) - c_{\text{reference}}(z) = q e^{-z/(1500 \text{ m})}$$

q is the unknown surface equivalent increase ($\mu\text{g}/\text{m}^3$). The profile shape is assumed, not independently recovered.

Vertical mass column: $(1500 \text{ m})q = 75,872.1 \mu\text{g}/\text{m}^2 = 75.8721 \text{ mg}/\text{m}^2$.

The reference is a baseline acquisition, not necessarily gas free air. Other species are assumed unchanged.

The fixed experiment used in all calculations

Parameter	Value
Geometry	550 km LEO satellite, 90° ground elevation
Spectrum	230–240 GHz, 1,024 tones; first tone 230.004883 GHz
Total RF power and apertures	25 dBm; 0.10 m transmit / 1.0 m receive; $\eta = 0.65$
Noise and additional loss	7 dB receiver noise figure; 5 dB implementation loss
Atmosphere and gas profile	ITU P.835 / P.676; CH ₃ CN scale height 1,500 m
Acquisition	10 s reference + 10 s sample; 0.3% full pilot symbols

The static benchmark assumes ideal tracking. No source demonstrates this entire orbital configuration.

Step 1 · Mass concentration becomes molecular density

$$n_1(0) = \frac{10^{-6} \text{ g/m}^3}{41.053 \text{ g/mol}} (6.02214076 \times 10^{23}) = 1.46691856 \times 10^{16} \text{ m}^{-3}$$

$$n(z; q) = q n_1(0) e^{-z/1500}$$

n_1 : density for a numerical concentration of 1 $\mu\text{g/m}^3$ · $n(z;q)$: molecular density for the chosen q
41.053 g/mol: natural mixture molar mass · q is inserted numerically in $\mu\text{g/m}^3$

At $q = 50.5814$, the surface molecular density is $7.419879\text{e}+17$ molecules/ m^3 .

HITRAN uses cm^2 per molecule, so the calculation converts density to cm^{-3} and path length to cm.

This converts an environmental concentration into the number of absorbers seen by the wave.

Step 2 · One actual HITRAN transition

Catalog parameter	Saved value	Physical role
Line center	239.1379167 GHz	Transition frequency
S(296 K)	2.616×10^{-21} cm/molecule	Integrated line strength
Air width γ_{air}	0.1594 cm ⁻¹ /atm	Collision broadening
Lower state energy E"	47.8643 cm ⁻¹	Temperature dependence
Width temperature exponent	0.71	Temperature scaling of width

This is one of 17,880 retained CH₃CN transitions. The calculation sums their contributions.

The measured tone is at 230.004883 GHz. A transition contributes away from its center through its broadened shape.

Step 2 · A line shape gives absorption at our tone

At $z = 16.8826$ m	Value
Atmospheric state	$T = 288.0403$ K; $p = 101,122.35$ Pa
Temperature corrected strength $S(T)$	2.845759×10^{-21} cm/molecule
Voigt value V at 230.004883 GHz	0.4334214 cm
Sum over all retained lines: $\sigma_o(z)$	2.642246×10^{-20} cm ² /molecule

$$\sigma_{\ell,0} = S_{\ell}(T)V_{\ell}(f_0; T, p) = (2.845759 \times 10^{-21})(0.4334214) = 1.233413 \times 10^{-21} \text{ cm}^2$$

$\sigma_{\ell,0}$: one transition's absorption cross section at tone 0 · $\sigma_o = \sum_{\ell} \sigma_{\ell,0}$

V is normalized over wavenumber. The temperature correction is worked out in the appendix.

σ is calculated from line parameters and atmospheric state. It is not a measured satellite attenuation.

Step 3 · The meaning of optical depth

Optical depth tells us how much absorption a wave accumulates along its path at a particular frequency.

Each thin layer contributes according to how many molecules it contains and how strongly they absorb. These contributions add along the path.

τ is dimensionless. The surviving power fraction is $\exp(-\tau)$. At $\tau = 1$, 36.8% survives and 63.2% is absorbed.

In this example, $\tau = 0.00280478$: the extra gas absorbs 0.2801% of the power, giving a 0.1401% amplitude reduction.

Here τ describes the additional CH_3CN absorption. Geometric spreading and background losses are separate.

Step 3 · One altitude node contributes optical depth

First quadrature node	Numerical value
Cross section σ_0	$2.6422462 \times 10^{-20} \text{ cm}^2/\text{molecule}$
Density $n_1(z)$ for $q = 1 \text{ } \mu\text{g}/\text{m}^3$	$1.4505008 \times 10^{10} \text{ molecules}/\text{cm}^3$
Integration weight w	$42.831123 \text{ m} = 4,283.1123 \text{ cm}$

$$\delta\tau_{0,1} = \sigma_0 n_1 w = (2.6422462 \times 10^{-20})(1.4505008 \times 10^{10})(4283.1123) = 1.6415372 \times 10^{-6}$$

$\delta\tau_{0,1}$: dimensionless optical depth contribution at tone 0 for $1 \text{ } \mu\text{g}/\text{m}^3$

The weight is a Gauss quadrature weight, not the thickness of a physical uniform slab.

Repeat this product at each altitude node and add the contributions.

Step 3 · Adding the atmosphere gives the tone sensitivity

Altitude interval	Optical depth for $q = 1 \mu\text{g}/\text{m}^3$	Attenuation sensitivity (dB per $\mu\text{g}/\text{m}^3$)
0–0.5 km	1.6389001e-05	7.1176528e-05
0.5–1 km	1.1583791e-05	5.0307766e-05
1–2 km	1.3936792e-05	6.0526719e-05
2–3 km	6.9036665e-06	2.9982243e-05
3–5 km	5.0671830e-06	2.2006496e-05
5–100 km	1.5704107e-06	6.8202070e-06

Sum: $\tau_{0,1} = 5.5450845 \times 10^{-5} \rightarrow a_0 = (10 / \ln 10) \tau_{0,1} = 0.00024081996 \text{ dB}/(\mu\text{g}/\text{m}^3)$

The vertical profile and atmospheric state determine this coefficient. It is not a universal CH_3CN constant.

Step 4 · The chosen concentration gives 0.012181 dB

$$\Delta A_0 = a_0 q = (0.0002408199585)(50.5813966) = 0.01218100983 \text{ dB}$$

$$\tau_0(q) = q\tau_{0,1} = (50.5813966)(5.545084465 \times 10^{-5}) = 0.002804781165$$

ΔA_0 : extra CH₃CN attenuation at the first tone · τ_0 : corresponding dimensionless optical depth

The 4.5477 dB oxygen/water loss belongs to the baseline link budget.

The 0.012181 dB value is the additional loss caused by this CH₃CN enhancement.

The concentration multiplies a sensitivity computed from real spectral parameters and an assumed path profile.

Step 5 · Absorption predicts the amplitude change

$$\frac{|h_{1,0}|}{|h_{0,0}|} = e^{-\tau_0/2} = 10^{-\Delta A_0/20} = 10^{-0.01218100983/20} = 0.9985985923$$
$$100 \left(1 - \frac{|h_{1,0}|}{|h_{0,0}|} \right) = 0.1401407692\%$$

$h_{0,0}$: reference channel at tone 0 · $h_{1,0}$: sample channel at tone 0, before measurement error

The ratio concerns channel magnitude after known geometric and instrument corrections.

The power ratio is 0.99719915. The stated 0.1401% is an amplitude reduction.

These are the third and fourth rows of the screenshot, now derived from the gas column.

A predicted change of 0.1401% is not yet evidence that a receiver can resolve it.

Step 6 · Observation time fixes the averaging gain

Resource	Calculation
Tone spacing and useful symbol	$10 \text{ GHz} / 1,024 = 9.765625 \text{ MHz}$; $T_u = 102.4 \text{ ns}$
Transmitted symbol	$102.4 \text{ ns} + 10 \text{ ns assumed CP} = 112.4 \text{ ns}$
Frame	10,000 symbols, with 30 full pilot symbols
Frames in each 10 s acquisition	$\text{floor}[10 / (10,000 \times 112.4 \text{ ns})] = 8,896$
Pilots per tone in each acquisition	$M = 8,896 \times 30 = 266,880$

Both the reference and sample are noisy: 10 s + 10 s, not a free or exact reference.

Ideal phase, frequency, delay and gain tracking is assumed throughout each moving acquisition.

Step 7 · One tone remains uncertain after averaging

First tone	Numerical value
Reference SNR per pilot, linear	$\rho_0 = 0.961035658$
Complex mean variance, normalized channel	$1 / (M\rho_0) = 3.8989213\text{e-}06$
Thermal variance of the reference/sample log ratio	$v_0(0) = 2.9415287\text{e-}04 \text{ dB}^2$
Total null variance after calibration term	$C_{00} = 2.9515287\text{e-}04 \text{ dB}^2$

$$v_0(0) = \frac{(20/\ln 10)^2}{2M} \left(\frac{1}{\rho_0} + \frac{1}{\rho_0} \right), \quad \sqrt{C_{00}} = 0.0171800 \text{ dB}$$

$C_{00} = v_0(0) + (0.001 \text{ dB})^2$ · The log ratio variance uses a high coherent mean SNR approximation.

20 s total · ideal tracking · assumed, unmeasured 0.001 dB differential calibration residual

Step 8 · The saved receiver draw does not equal its mean

Normalized complex pilot means, simulated	Value
Reference $\hat{h}_{0,0}$	$1.0009162403 + j\ 0.0004400251$
Sample $\hat{h}_{1,0}$	$1.0007326946 - j\ 0.0009412581$
Magnitudes $ \hat{h}_{0,0} $ and $ \hat{h}_{1,0} $	1.0009163371 and 1.0007331373
Persistent differential calibration draw e_0	-0.0020537566 dB

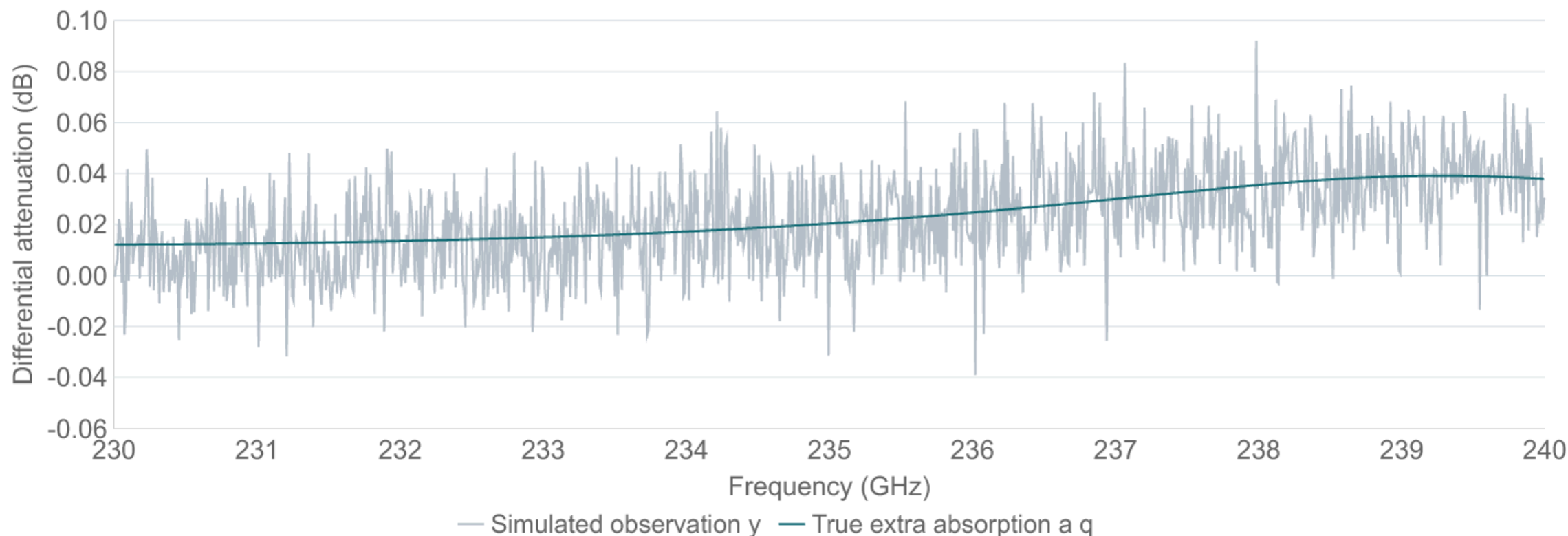
$$y_0 = -20 \log_{10} \left(\frac{1.0007331373}{1.0009163371} \right) - 0.0020537566 = 0.0015899416 - 0.0020537566 = -0.0004638149 \text{ dB}$$

The mean sample amplitude is 0.9985985923, but the noisy pilot mean can be larger. Seed: 20261005.

A positive gas enhancement can produce a negative observed attenuation at one noisy tone.

These are simulated complex observations with the stated noise and calibration model, not field measurements.

Step 9 · The detector uses the complete observed spectrum



Input to estimation: y has 1,024 entries. The first is -0.0004638 dB, not the predicted $+0.012181$ dB.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Step 10 · The design matrix encodes gas and gain changes

Tone index k	$a_k [10^{-4} \text{ dB}/(\mu\text{g}/\text{m}^3)]$	Offset column	Slope u_k
0	2.408200	1	-0.500000
255	2.804245	1	-0.250733
511	4.032712	1	-0.000489
767	6.489083	1	0.249756
1023	7.500802	1	0.500000

$$\mathbf{y} = \mathbf{A}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \mathbf{A} = [\mathbf{a} \ \mathbf{1} \ \mathbf{u}] \in \mathbb{R}^{1024 \times 3}, \quad \boldsymbol{\theta} = [q \ b_0 \ b_1]^T$$

Five displayed rows are selected from all 1,024. $u_k = (f_k - \text{mean}(f)) / (\text{max}(f) - \text{min}(f))$.

b_0 : common gain change (dB) · b_1 : spectral gain slope coefficient (dB) · ε : remaining observation error (dB)

A contains known model responses. The unknowns are q and the two gain coefficients.

Step 11 · The covariance encodes which errors average down

$$\mathbf{C}(q) = \text{diag}(v_k(q)) + 10^{-6}\mathbf{R}, \quad R_{ij} = e^{-|f_i - f_j|/(10 \text{ GHz})}$$
$$v_k(q) = \frac{(20/\ln 10)^2}{2M} [\rho_{0,k}^{-1} + \rho_{1,k}(q)^{-1}], \quad \rho_{1,k}(q) = \rho_{0,k} 10^{-a_k q/10}$$

C is $1,024 \times 1,024$ in dB². Its thermal term includes reference and sample noise.
 10^{-6} dB² is the assumed differential calibration variance; it is not divided by M.

For example: $C_{00} = 2.95152865 \times 10^{-4}$ dB², while $C_{0,1023} = 3.68238874 \times 10^{-7}$ dB².

The estimator fixes its weights using C(0). Detection probability at $q > 0$ uses the corresponding C(q).

The calibration covariance is an explicit unmeasured assumption, not an estimate obtained from these data.

Step 12 · Weighted regression estimates the unknowns

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} (\mathbf{y} - \mathbf{A}\boldsymbol{\theta})^T \mathbf{C}(0)^{-1} (\mathbf{y} - \mathbf{A}\boldsymbol{\theta}), \quad \mathbf{G}\hat{\boldsymbol{\theta}} = \mathbf{g}$$

$\mathbf{G} = \mathbf{A}^T \mathbf{C}(0)^{-1} \mathbf{A}$, a 3×3 matrix · $\mathbf{g} = \mathbf{A}^T \mathbf{C}(0)^{-1} \mathbf{y}$, a 3×1 vector

$\boldsymbol{\theta} = [\hat{q}, \hat{b}_0, \hat{b}_1]^T$ · $\mathbf{C}(0)^{-1}$ accounts for unequal noise and cross frequency correlation

The fit seeks the concentration whose spectral shape best explains y after allowing gain offset and slope.

The next slide gives every entry of the resulting three dimensional system.

The receiver does not divide each observed tone by a_k and average the answers.

Step 12 · The full spectrum reduces to this numerical system

Equation row	Coefficient of q	Coefficient of b ₀	Coefficient of b ₁	Right side g
1	0.301344020	458.463514	127.082723	15.066265191
2	458.463513671	983654.579709	-223.779287	22141.401031152
3	127.082722926	-223.779287	198502.414349	7023.360955313

$$\begin{bmatrix} G_{qq} & \mathbf{G}_{qb} \\ \mathbf{G}_{bq} & \mathbf{G}_{bb} \end{bmatrix} \begin{bmatrix} \hat{q} \\ \hat{\mathbf{b}} \end{bmatrix} = \begin{bmatrix} g_q \\ \mathbf{g}_b \end{bmatrix}$$

The matrix columns correspond to different parameter units: q in µg/m³, b₀ and b₁ in dB.
All values use all 1,024 tones. Displayed coefficients are rounded; source arrays retain full precision.

The small q coefficient alone is not the usable information: much of the gas pattern resembles gain changes.

Step 13 · Eliminating gain terms gives the concentration

$$I_q = G_{qq} - \mathbf{G}_{qb} \mathbf{G}_{bb}^{-1} \mathbf{G}_{bq} = 0.3013440205 - 0.2951744347 = 0.006169585805$$

$$t_q = g_q - \mathbf{G}_{qb} \mathbf{G}_{bb}^{-1} \mathbf{g}_b = 15.0662651908 - 14.8230312721 = 0.243233918673$$

$$\hat{q} = t_q / I_q = 39.42467555 \text{ } \mu\text{g}/\text{m}^3$$

I_q : information for q after removing the gain nuisance terms $[(\mu\text{g}/\text{m}^3)^{-2}]$

t_q : remaining weighted evidence for q $[(\mu\text{g}/\text{m}^3)^{-1}]$

The same solve gives $\hat{b}_0 = 0.00413651 \text{ dB}$ and $\hat{b}_1 = 0.01014643 \text{ dB}$.

The true simulated q was $50.5814 \text{ } \mu\text{g}/\text{m}^3$. The estimate is different because this observation contains error.

20 s total · ideal tracking · assumed, unmeasured 0.001 dB differential calibration residual

Step 14 · The estimate exceeds a prespecified threshold

$$s_0 = I_q^{-1/2} = 12.73127783 \mu\text{g}/\text{m}^3$$

$$q_{\text{th}} = z_{0.99}s_0 = (2.326347874)(12.73127783) = 29.61738111 \mu\text{g}/\text{m}^3$$

$$\hat{q} = 39.42467555 > 29.61738111 \implies \text{detection}$$

H_0 : $q = 0$ · H_1 : $q > 0$ · s_0 : standard deviation under H_0

q_{th} : threshold for 1% false alarm probability for this one prespecified compound and decision

The decision says the observed pattern is inconsistent with no enhancement under this model.

It does not establish exact concentration, unique chemical identity in a changing mixture, or field performance.

20 s total · ideal tracking · assumed, unmeasured 0.001 dB differential calibration residual

Why the chosen q was $50.5814 \mu\text{g}/\text{m}^3$

$$\Pr(\hat{q} > q_{\text{th}} \mid q) = 0.95 \iff q - q_{\text{th}} = z_{0.95} s_1(q)$$

$$q_{95} = 29.61738111 + (1.644853627)(12.74521644) = 50.58139660 \mu\text{g}/\text{m}^3$$

$s_1(q) = \sqrt{[H C(q) H^T] \cdot H}$ is the first row of $G^{-1} A^T C(0)^{-1}$, so $\hat{q} = H y$.
 $z_{0.95}$: standard normal 95th percentile · these weights are fixed under the null.

This concentration was selected from the analytical sensitivity calculation before the random receiver draw.

It is a controlled simulation input. It was neither measured in the atmosphere nor estimated from the sample.

The threshold is $29.6174 \mu\text{g}/\text{m}^3$. Achieving 95% predicted response requires a larger true enhancement.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Repeated observations show success and failure rates

10,000 simulated reference/sample pairs at each concentration, using the same fixed threshold.

True q ($\mu\text{g}/\text{m}^3$)	Predicted response	Simulated response	95% interval
0	1.00%	1.13%	0.93–1.36%
1	1.23%	1.31%	1.10–1.55%
50.5814	95.00%	94.75%	94.29–95.18%

1 $\mu\text{g}/\text{m}^3$ is not reliably detected: the simulated response is only 1.31%, near the false alarm rate.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

What is real, assumed and simulated in this example

Evidence category	What belongs to it
External physical inputs	Actual HITRAN records; published ITU atmospheric models
Design assumptions	550 km geometry; hardware configuration; exponential gas enhancement
Unmeasured error assumption	0.001 dB differential residual and its spectral covariance
Simulated observation	Complex pilot means, calibration draw and resulting y
Not demonstrated	Measured atmospheric detection, ambient sensitivity or 3D reconstruction

The calculation is concrete and reproducible. It is not an orbital measurement.

A selected aircraft background reference is about $0.252 \mu\text{g}/\text{m}^3$, far below this conditional enhancement limit.

The original table, now with its calculation attached

Quantity	How the value is obtained	Result
Chosen enhancement	Solve the 95% power equation, then set the simulation input	50.5814 $\mu\text{g}/\text{m}^3$
Extra absorption at tone 0	$(0.0002408199585) \times 50.5813966$	0.012181 dB
Sample/reference amplitude	$10^{(-0.01218100983 / 20)}$	0.998599
Amplitude reduction	$100 \times (1 - 0.9985985923)$	0.1401%

After adding the modeled errors and fitting all tones: $\hat{q} = 39.4247 \mu\text{g}/\text{m}^3 > \text{threshold } 29.6174 \rightarrow \text{detection.}$

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Appendix · Temperature correction for the actual line

$$S(T) = S(296) \frac{Q(296)}{Q(T)} \exp \left[-c_2 E'' \left(\frac{1}{T} - \frac{1}{296} \right) \right] \frac{1 - e^{-c_2 \tilde{\nu}/T}}{1 - e^{-c_2 \tilde{\nu}/296}}$$

$Q(T)$: molecular partition sum · $E'' = 47.8643 \text{ cm}^{-1}$ · $\tilde{\nu} = 7.97678228 \text{ cm}^{-1}$

c_2 : second radiation constant (cm K) · $T = 288.040263 \text{ K}$

$$Q(296) / Q(T) = 88658.97016 / 83172.06723 = 1.065970501.$$

$$S(T) = 2.616 \times 10^{-21} \times 1.065970501 \times 0.993591393 \times 1.027087331 = 2.845759 \times 10^{-21} \text{ cm/molecule.}$$

The line strength changes with state populations and stimulated emission before the profile is applied.

Appendix · Power and noise at the first tone

Link budget term	Value
Transmit power per tone	$25 - 10 \log_{10}(1024) = -5.1030 \text{ dBm}$
Transmit + receive gains	$45.7705 + 65.7705 \text{ dBi}$
Free space + background + implementation loss	$194.4898 + 4.5477 + 5.0000 \text{ dB}$
Reference / sample received power	$-97.5996 / -97.6118 \text{ dBm}$
Noise power in one tone	-97.4270 dBm
Reference per pilot SNR	$-0.172605 \text{ dB} = 0.96103566 \text{ linear}$

The extra 0.012181 dB CH₃CN loss belongs only to the sample in this enhancement model.

Appendix · Five actual rows of the covariance

This is a submatrix of the full null covariance, not a new error model.

C / 10 ⁻⁴ dB ²	0	255	511	767	1023
0	2.951529	0.007796	0.006071	0.004728	0.003682
255	0.007796	2.947224	0.007788	0.006065	0.004724
511	0.006071	0.007788	2.947702	0.007788	0.006065
767	0.004728	0.006065	0.007788	2.952581	0.007788
1023	0.003682	0.004724	0.006065	0.007788	2.961572

The off diagonal elements carry the assumed correlation between spectral errors.

20 s total • ideal tracking • assumed, unmeasured 0.001 dB differential calibration residual

Appendix · A five tone dot product can be checked by hand

Tone k	Observed y_k (dB)	Five tone weight $H_{5,k}$	Product $H_{5,k} y_k$
0	-0.000463815	4942.686	-2.292492
255	0.025099503	-4201.176	-105.447434
511	-0.031417175	-5681.920	178.509866
767	0.018701897	4215.931	78.845915
1023	0.030551022	724.479	22.133560

Sum = 171.7494 $\mu\text{g}/\text{m}^3$; standard deviation = 165.1163; threshold = 384.1179 → no detection.

This discards 1,019 tones. The main detector uses all 1,024, with the same assumed 0.001 dB residual and 20 s total.

Appendix · Hardware evidence keeps its original scope

Source	Useful evidence	What it does not establish
Sen et al., Nature Electronics	200 mW at 210–240 GHz, terrestrial link	Orbital operation
TeraLink, 2026 preprint	25 dBm, 7 dB NF, proposed LEO system	Flight validation of this example
Cooper et al., JMW	>400 mW CW at 240 GHz	A complete 10 GHz OFDM modem

The selected parameters have physical references, but combined bandwidth, calibration and tracking remain unverified.

1 W and 10 W transmit cases elsewhere in the repository are sensitivity scenarios, not qualified payload specifications.

Sources · Spectroscopy and atmospheric state

HITRAN: actual CH₃CN line records, units, temperature correction and absorption model.
hitran.org/docs/definitions-and-units/

HAPI and TIPS2025: partition functions used in the spectral calculation. Kochanov et al., JQSRT 177 (2016), DOI 10.1016/j.jqsrt.2016.03.005.

ITU P.835-7 (2024): standard atmospheric state. ITU P.676-13 (2022): oxygen/water propagation and emission.

EMeRGe / IAGOS context: ACP 23, 1893 (2023), DOI 10.5194/acp-23-1893-2023.

Sources · Every numerical step is saved

`worked_steps.json` contains the values substituted in the equations.

`quadrature_first_tone.csv` and `altitude_contributions.csv` expose the complete atmospheric integral.

`complex_receiver_steps.csv` records both complex pilot means and the final observation at every tone.

`normal_equations.npz` stores G , g and θ ; the original `matrices.npz` contains the full A , C and H .

Reproduce with `scripts/calculate_zenith_worked_steps.py`. The saved source hashes are in `manifest.json`.